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DTS 211 — Past Questions & Materials

by Renance

Course: DTS 211

Institution: Federal University of Technology, Akure

Sources in this pack: 1

Includes past questions and lecture notes

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1. Perform $15 + 8 / 2$. Show R's order of operations.

R follows **BODMAS**..

$$15 + 8 / 2 = 19$$

1i. Rewrite the expression with parentheses to change the result.

$$(15 + 8) / 2 = 11.5$$

2. How do you compute exponents in R?

Exponents are computed in R using “^”

2i. Compute 7^3 using R.

$7^3 = 343$

3. What is the modulus operator in R?

%% is the modulus operator

3i. Use it to find the remainder of $29 \div 5$.

`29 %% 5` =4

4. Explain vectorized arithmetic in R

R applies operations element-wise.

4i. Add two vectors: $c(1,2,3) + c(4,5,6)$.

$c(1,2,3) + c(4,5,6) = 5\ 7\ 9$

5. How does R handle different-length vectors in operations?

By recycling shorter vectors to match with the longer ones.

5i. Demonstrate with: $c(1,2,3,4) + c(10,20)$.

$c(1,2,3,4) + c(10,20) = (1+10, 2+20, 3+10, 4+20) = 11\ 22\ 13\ 24$

6. What is integer division in R?

This returns the **whole-number quotient** of a division, ignoring any remainder. It uses the `%/%` operator.

6i. Compute `29 %/% 4`.

```
29 %/% 4 = 7
```

7. How do you calculate the square root of a number?

By using the `sqrt()` function.

7i. Calculate `sqrt(144)`.

`sqrt(144)` = 12

8. What function do you use to compute logarithms?

You compute logarithms in R using the `log()` function.

8i. Compute $\log_{10}(1000)$.

$\log_{10}(1000) = 3$

9. Explain the difference between `sum()` and `cumsum()`.

`sum()` gives you one final total of all the values in a vector, while `cumsum()` gives you a running total that shows how the sum grows at each step.

9i. Apply both to the vector `c(2,4,6)`.

```
sum(c(2,4,6)) # 12
```

```
cumsum(c(2,4,6)) # 2 6 12
```

10. What is the purpose of the round() function?

The round() function is used for rounding up values to a certain point

10i. Round 3.6789 to 2 decimal places

3.68

11. How do you generate sequences of numbers?

You generate sequences of numbers in R using the `:` operator for simple increments or the `seq()` function when you need more control over steps or length.

11i. Generate numbers from 5 to 50 in steps of 5

```
seq(5, 50, by=5) # 5 10 15 ... 50
```

12. What does mean() compute in R?

It is for computing the average of given values

12i. Compute mean of c(4,6,8,10).

`mean(c(4,6,8,10)) = 7`

13. How do you compute variance in R?

You compute variance in R using the `var()` function

13i. Also compute standard deviation of the same vector.

```
var(c(4,6,8,10)) = 6.666667
```

```
sd(c(4,6,8,10)) = 2.581989
```

14. What does the abs() function do?

The abs() function returns the **absolute value** of a number

14i. Apply abs() to the vector c(-3,4,-9).

```
abs(c(-3,4,-9)) # 3 4 9
```

15. Explain recycling in arithmetic operations.

In arithmetic operations, recycling in R means that when vectors of different lengths are combined, the shorter vector's elements are repeated (recycled) to match the length of the longer vector.

15i. Multiply `c(2,4,6)` by `c(3)`.

```
c(2,4,6) * c(3) = 6 12 18
```

16. What is the function for cumulative product?

The function for cumulative product in R is `cumprod()`.

16i. Apply it to `c(1,2,3,4)`.

```
cumprod(c(1,2,3,4)) # 1 2 6 24
```

17. How do you check if a number is even or odd?

You check if a number is even or odd in R using the modulo operator `%%`

17i. Write a short expression that checks if 17 is odd

```
17 %% 2 != 0 # TRUE → odd
```

18. What does the range() function return?

The range() function returns a vector containing the **minimum and maximum** values of its input.

18i. Use it on c(5,10,3,20).

```
range(c(5,10,3,20)) # 3 20
```

19. What is the difference between min() and which.min()?

min() gives you the smallest value in a vector, whereas which.min() gives you the **position** (index) of that smallest value.

19i. Apply both on c(10, 4, 7, 4).

```
min(c(10,4,7,4))    # 4
```

```
which.min(c(10,4,7,4)) # 2
```

20. How do you perform matrix multiplication in R?

In R, you perform matrix multiplication using the `%*%` operator, which multiplies two matrices according to the rules of linear algebra.

20i. Multiply two 2×2 matrices.

```
A <- matrix(c(1,2,3,4), nrow=2)
```

```
B <- matrix(c(5,6,7,8), nrow=2)
```

```
A %*% B
```

```
# [,1] [,2]
```

```
# [1,] 19 22
```

```
# [2,] 43 50
```